

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

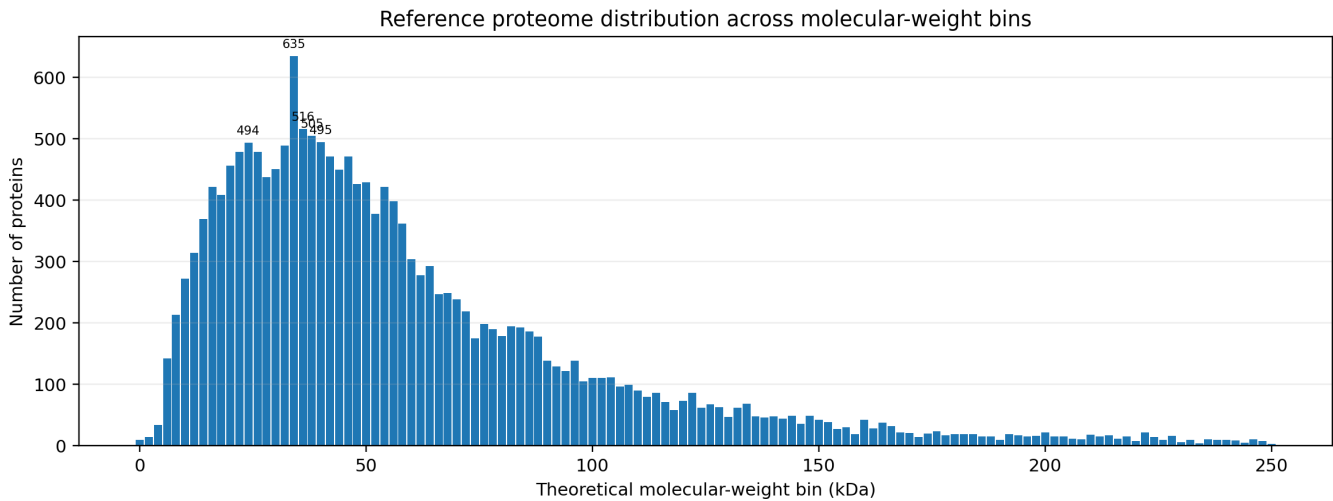
Metric	Value
Proteins parsed	18,372
Molecular-weight bins	250
Bin width	2 kDa
MW range	0.64 to 3895.53 kDa
Median MW	48.70 kDa
Mean MW	65.12 kDa
Proteins <= 50 kDa	9,454 (51.5%)
Proteins 50-100 kDa	5,946 (32.4%)
Proteins > 100 kDa	2,972 (16.2%)
High-priority bins	25
Elevated-priority bins	38
Routine-priority bins	187
Source FASTA	chicken_UP000000539_9031.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

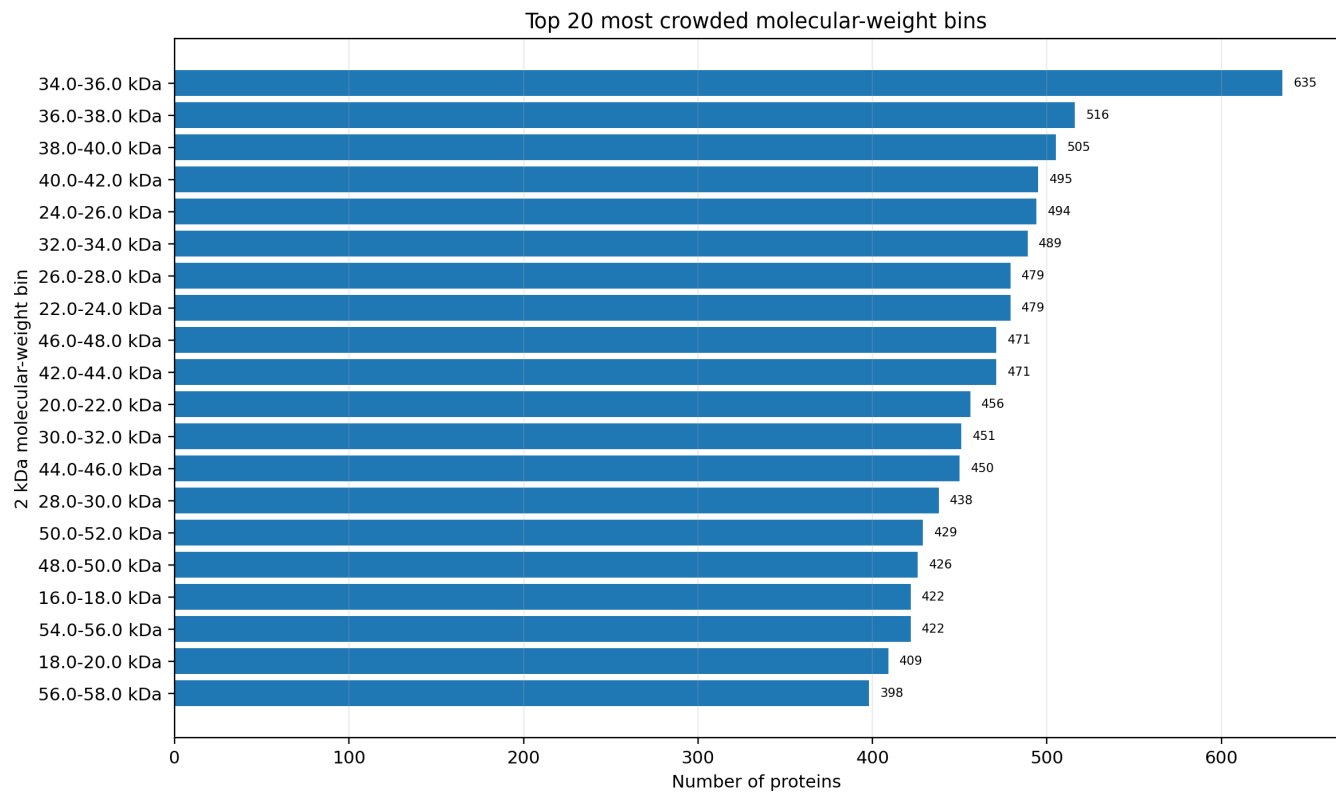
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

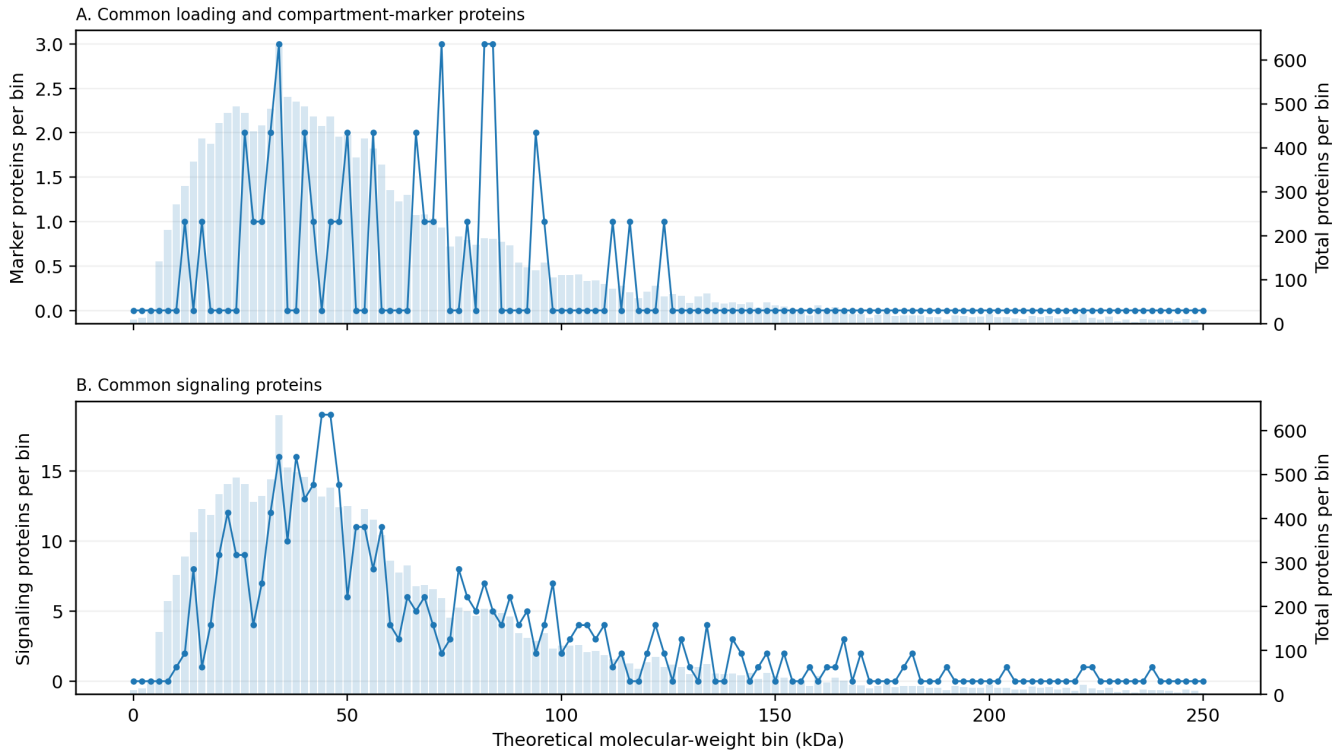
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

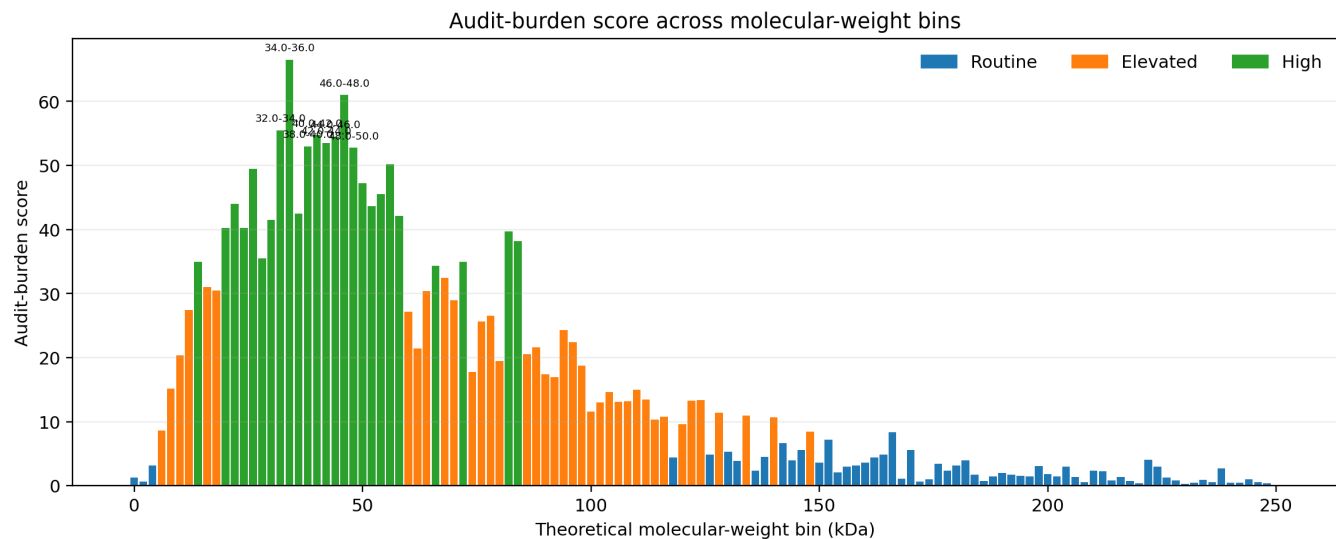
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
34.0-36.0 kDa	635	3	16	high	66.5
46.0-48.0 kDa	471	1	19	high	61.0
32.0-34.0 kDa	489	2	12	high	55.5
40.0-42.0 kDa	495	2	13	high	54.8
44.0-46.0 kDa	450	0	19	high	54.5
42.0-44.0 kDa	471	1	14	high	53.5
38.0-40.0 kDa	505	0	16	high	53.0
48.0-50.0 kDa	426	1	14	high	52.8
56.0-58.0 kDa	398	2	8	high	50.1
26.0-28.0 kDa	479	2	9	high	49.5
50.0-52.0 kDa	429	2	6	high	47.2
54.0-56.0 kDa	422	0	11	high	45.5
22.0-24.0 kDa	479	0	12	high	44.0
52.0-54.0 kDa	378	0	11	high	43.6
36.0-38.0 kDa	516	0	10	high	42.5
58.0-60.0 kDa	362	0	11	high	42.1
30.0-32.0 kDa	451	1	7	high	41.5
24.0-26.0 kDa	494	0	9	high	40.2
20.0-22.0 kDa	456	0	9	high	40.2
82.0-84.0 kDa	195	3	7	high	39.8
84.0-86.0 kDa	193	3	5	high	38.1
28.0-30.0 kDa	438	1	4	high	35.5
14.0-16.0 kDa	369	0	8	high	35.0
72.0-74.0 kDa	219	3	2	high	35.0
66.0-68.0 kDa	247	2	5	high	34.4

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
34.0-36.0 kDa	635	3	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UNG, MADPRT, SLC35E3, RASSF3, ORC6, , RAX, SPDEF, CD200R1A, REEP6, FAM168A, SIAH2, TMEM177, RX2, H0XC11, SPARC, SULT1B1, PPIL6
46.0-48.0 kDa	471	1	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	NTSR1, LIMS1, GLA, SS18, SCRN1, ABRAXAS1, BMP4, CZH18ORF25, C17orf58, CCNA1 [Cell-cycle CDK/cyclin], AADACL3B, SPATA17, PKDCCB, NR2F1, KRT19, ALDH1A2, KDF1, PIP4K2A
32.0-34.0 kDa	489	2	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, HUS1, , RSPO3, AQP3, EMCN, NFU1 [Mitochondrial markers], ANKRD39, JAML, EXOSC8, KRT222, MMACHC [Lamin family], THEM4, LOC415758, NSL1, KCNM2B, MNR2, CENPQ
40.0-42.0 kDa	495	2	13	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; p53 pathway	high	PARP11, SH3GL3, , CFAP410, IL10RB, FAM45A, SH3GL2, RPLP0, GNA2, WNT8C, NECAB3, PMEPA1, SETD7 [Histone family], GLT8D2, PEX16, RABEPK, TNFRSF9, ATP6V0D1
44.0-46.0 kDa	450	0	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers	high	MAPK9 [MAPK cascade], RPRD1B, SERPINE3, SEH1L, SERPINE2, OXTR, E2F1, LMX1B, LOC121109782, ONECUT2, RBM41, MAP2K2 [MAPK cascade], RPL3, SCEL, CCNA2 [Cell-cycle CDK/cyclin], CCDC94, MRPS35 [Mitochondrial markers], LOC107049146
42.0-44.0 kDa	471	1	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	PRSS23, BGV, LOC100858953, ACTBL2 [Actin family], NR1I3, B3GNT3, ACTC1 [Actin family], TPST2, SGK494, ACTB [Actin family], HSD11B2, LYAR, B3GNT10, TPST2, LZTFL1, ACTA1 [Actin family], PGA5, GNA11
38.0-40.0 kDa	505	0	16	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, NODAL, SNX20, RNF215, ABHD6, HSF2BP [Heat shock proteins], SORD, DUSP23, CD8A, LOC101751271, AHSA1 [Heat shock proteins], GALT, TNFRSF6B, EXTL2, BMERB1, HAGH [Mitochondrial markers], RNF146, TIMP2
48.0-50.0 kDa	426	1	14	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	high	KMT5A [Histone family], GLUL, TTL9 [Tubulin family], MTHFSO, KCTD18, BZW1, DRD3, SYT13, ABHD2, BMP6, TBX19, JMJD6, LHX2, FFAR2L7, RCC1L, MHCY15, AP4M1, HS6ST1
56.0-58.0 kDa	398	2	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	TMEM39B, SH2D2A, ALDH3B1, , LRRRC71, DUS1L, CHRN2B, ENTDP8, , EDC3, KATNA1, CHRN84, PCYOX1, WASF3L, DFNA5, GABRB4, TMEM161B, CREB5
26.0-28.0 kDa	479	2	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RABL3, TSN, , PPP1R2, COQ10B [Mitochondrial markers], RAB3C, , PRSS3, C1QC, , LOC107049069, TXNL4A, KLK7, LOC768817, SCAMP4, , EFNA5, GCH1
50.0-52.0 kDa	429	2	6	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TUBB6 [Tubulin family], AZIN1, DLST [Mitochondrial markers], CHRNA10, MTERF2 [Mitochondrial markers], FAM188B2, NAF1, CHRNA10, CELF5, MFSO9, TUBA3E [Tubulin family], C4BPM, , NPY5R, PTAR1, C7orf26, CADM1, SERPINF1
54.0-56.0 kDa	422	0	11	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	TRMT13, IL17REL, HSPA13 [Heat shock proteins], SYT17, PLEKHD1, THUMP2D, HSPA14 [Heat shock proteins], MATN1, ENTDP8, SKA3, EVA1C, G3BP2, AP5M1, ATF2, MAPKAPK5 [MAPK cascade], CAMK2A, SLC2A1, HEATR8L5
22.0-24.0 kDa	479	0	12	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	, VSTM2L [Histone family], CLEC3A, TRNP1, DNAJC5B, COX5B [Mitochondrial markers], [Histone family], SNX24, C8FB, FAM219B, LOC121106526, LOC121108905, , BMF, LOC100858177, CD320, MRPS23
52.0-54.0 kDa	378	0	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	high	PRR33, CRBN, CPQ, PTGER4, PPP2R2D, GLRA2, FAM134B, SUSDA4, F9, GFRA1, , HARB1, PLD3, GPBP1L1, GDF10, TSEN2, NTS5DC1, CARNMT1
36.0-38.0 kDa	516	0	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	CHIR3B3, LOC112533538, MTERF4 [Mitochondrial markers], AASDHPPT, TMEM189, IRF1, PPP1R3C, TOB1, LOC121112382, APTX, NEIL2, ZFP36L1, STAR [Mitochondrial markers], LRRCC23, TNFAIP1, , C14ORF166B, CIGALT1C1
58.0-60.0 kDa	362	0	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	GJA10, LACTB, PKM, KPNA2, LIPI, RFX5, , SESN3, , ALDH7A1, IMPDH2, SNTG1, SLC68A1, SEPTIN4, CASP8 [Apoptosis caspase/BCL], CYP19A1, SLC16A8, MIER1
30.0-32.0 kDa	451	1	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, ITM2B, TMEM220, DRAM2, HOXD12, TFAM [Mitochondrial markers], ELOVL3, SNAI2, ZFP511, AGPAT2, , LOC101747821, MYOZ2, PENK, UPRT, IGF8P5, TMEM106C, MRPL19
24.0-26.0 kDa	494	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	, TCTEX1D4, PSDM10, CHMP2B, CCDC59, TMEM140, RAB4A, PXMP4, FZD8, , TCEANC2, RPS8, THPO, CHMP2B, , LOC124416945, RAB19, VPS29L
20.0-22.0 kDa	456	0	9	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	, FAM180A, FAM168B, CRYAB, , LOC121109670, TMEM47, MPLKIP [Actin family], LOC121108888 [Mitochondrial markers], , TRAPPC3, , TIMM17B [Mitochondrial markers], MRPS25, LOC768589, C5orf24, LOC100858647, ARF6
82.0-84.0 kDa	195	3	7	Actin family; Heat shock proteins; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	CEP63, ANKRD13D, VRTN, SMYD4, USP49, SLC26A6, AK7, PAPOLG, XPR1, CEP63, IGSF21, ESPN, CNOT10, ACSS2, COLGALT2, POMT2, IMMT, EXT2
84.0-86.0 kDa	193	3	5	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	high	MTSS1L, SGIP1 [Actin family], NCKIPSD [Actin family], HSP90AA1 [Heat shock proteins], KIF2A, LRFN5, RUBCNL, SMURF1, EXOC8, KRIT1, HMNR, TSGA10, HSP90AA1 [Heat shock proteins], RPS6KA6, TRDN, FCHSD2, ZNF770, TMEM8A
28.0-30.0 kDa	438	1	4	Actin family; Cell-cycle CDK/cyclin; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	RCHY1, KIAA1644, UTP23, FOLR3, RPL8, , ZCCHC9, , PEX11B, PLPP5, DOLPP1, LOC100859906, SRSF1, MVB12A, , CLTB, PSMA7, SCO2 [Mitochondrial markers]
14.0-16.0 kDa	369	0	8	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	FAM103A1, H2A-IX [Histone family], IMPP2L [Mitochondrial markers], , , NDUFS6 [Mitochondrial markers], , , , , CCK, LOC121107574, UTS2, ISCA1 [Mitochondrial markers], TAC1, LSM10, TMEM35B
72.0-74.0 kDa	219	3	2	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	TMEM62, SLC6A14, HSPA5 [Heat shock proteins], MTMR6, FRMD1, RAD21, OTUD5, , PLPPR3, TRIM7.1, PIAS3, MED17, DDX3X, FBXL16, PIGR, , LOC121107539, FLAD1
66.0-68.0 kDa	247	2	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	RAP1GDS1, BFSP2, KLHL10, ENC1, PCTHD3, FBXW7, P2RX7, SLC2A1, CTPS2, GPR107, , CRY2, KLHL7, NEIL3, BTBD17, PLEKHO2, AIFM3, KLHL21
68.0-70.0 kDa	249	1	6	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	RPA1, PTPN9, LOC419409, SLC18A2, KLHL20, COLGALT1, ANKRD13A, LOC121108224, GALTN9, ABCD4, PRKCI, SPATA7, LOC429032, PAPSS2, CERT1, CANX [ER markers], NTN1, SGO1
16.0-18.0 kDa	422	1	1	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; p53 pathway	elevated	, LITAF, , PNRC2, , , UMAD1, MCFD2, LOC121108846, LIN52, GNG4, RPL22, CATHL1, PNRC2, LOC112531842, C19orf70, IL18BP
18.0-20.0 kDa	409	0	4	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family	elevated	MMGT1, UBE2I, GCSH [Mitochondrial markers], , SFT2D1, , CPLX4, PPP1R17, NGB, PLP2, , PPIL3, TCTEX1D2, LOC107050964 [Mitochondrial markers], SAMD12, , NUDT1, UBALD1
64.0-66.0 kDa	293	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	GPC5, IL2RB, UBQLN4, CNOT6, WFIKN2, LOC107056275, GALTN1, HABP2, OPTN, A1CF, ASNS, PRKAA1, FANCC, SLAIN2, BPLI3, IL12RB1, , SLC27A2
70.0-72.0 kDa	239	1	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	ENTPD1, ZBTB44, CROT, LOC420419, RUFY2, TRAF4, ZBTB7C, ICK, PODXL2, XPNPEP1, NEURL1, SLCO2A1, ASIC4, EYAA, TNIP1 [Actin family], KLHL40, KRT3, KBTBD3

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
13.58	H2AFZ	Q5ZMD6	H2AZ_CHICK	Histone H2A.Z	12.0-14.0 kDa
16.42	TOMM20	A0A1D5PKT3	A0A1D5PKT3_CHICK	Translocase of outer mitochondrial membrane 20	16.0-18.0 kDa
27.54	PCNA	A0A8V0ZEF6	A0A8V0ZEF6_CHICK	DNA sliding clamp PCNA	26.0-28.0 kDa
27.63	COX4I1	A0A8V0XW68	A0A8V0XW68_CHICK	Cytochrome c oxidase subunit 4	26.0-28.0 kDa
28.89	PCNA	Q9DEA3	PCNA_CHICK	DNA sliding clamp PCNA	28.0-30.0 kDa
30.71	VDAC1	E1BYN7	E1BYN7_CHICK	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
33.13	TBP	O13270	TBP_CHICK	TATA-box-binding protein	32.0-34.0 kDa
33.18	TBP	Q6JLB1	Q6JLB1_CHICK	TATA-box binding protein	32.0-34.0 kDa
34.29	RPLP0	P47826	RLA0_CHICK	Large ribosomal subunit protein uL10	34.0-36.0 kDa
35.69	VDAC2	A0A8V0ZPZ5	A0A8V0ZPZ5_CHICK	Non-selective voltage-gated ion channel VDAC2	34.0-36.0 kDa
35.70	GAPDH	P00356	G3P_CHICK	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
40.02	RPLP0	A0A8V0ZQJ3	A0A8V0ZQJ3_CHICK	Large ribosomal subunit protein uL10	40.0-42.0 kDa
41.74	ACTB	P60706	ACTB_CHICK	Actin, cytoplasmic 1	40.0-42.0 kDa
42.04	ACTB	A0A8V0ZIN3	A0A8V0ZIN3_CHICK	Actin, beta	42.0-44.0 kDa
47.62	TUBB2A	A0A8V0YNJ4	A0A8V0YNJ4_CHICK	Tubulin beta chain	46.0-48.0 kDa
48.33	CALR	A0A8V0X980	A0A8V0X980_CHICK	Calreticulin	48.0-50.0 kDa
50.14	TUBA1A	P02552	TBA1A_CHICK	Tubulin alpha-1A chain	50.0-52.0 kDa
50.41	TUBB3	F1NMU4	F1NMU4_CHICK	Tubulin beta chain	50.0-52.0 kDa
57.41	P4HB	P09102	PDIA1_CHICK	Protein disulfide-isomerase	56.0-58.0 kDa
57.63	P4HB	A0A8V1AL50	A0A8V1AL50_CHICK	Protein disulfide-isomerase	56.0-58.0 kDa
66.53	LMNB1	P14731	LMNB1_CHICK	Lamin-B1	66.0-68.0 kDa
66.55	LMNB1	A0A8V0YN43	A0A8V0YN43_CHICK	Lamin-B1	66.0-68.0 kDa
68.13	CANX	A0A8V0YS98	A0A8V0YS98_CHICK	Calnexin	68.0-70.0 kDa
70.83	HSPA8	O73885	HSP7C_CHICK	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.93	SDHA	Q9YHT1	SDHA_CHICK	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
72.94	SDHA	A0A8V0XRG9	A0A8V0XRG9_CHICK	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
73.16	LMNA	P13648	LMNA_CHICK	Lamin-A	72.0-74.0 kDa
78.05	HSPA8	A0A8V1A5Y2	A0A8V1A5Y2_CHICK	Heat shock protein family A (Hsp70) member 8	78.0-80.0 kDa
82.55	HSP90AB1	A0A8V0ZAR1	A0A8V0ZAR1_CHICK	Heat shock protein 90 alpha family class B member 1	82.0-84.0 kDa
82.63	TUBA1A	A0A8V0ZZM1	A0A8V0ZZM1_CHICK	Tubulin alpha chain-like 3	82.0-84.0 kDa
83.43	HSP90AB1	Q04619	HS90B_CHICK	Heat shock cognate protein HSP 90-beta	82.0-84.0 kDa
84.06	HSP90AA1	P11501	HS90A_CHICK	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
84.12	HSP90AA1	A0A1D5PUL6	A0A1D5PUL6_CHICK	Heat shock protein 90 alpha family class A member 1	84.0-86.0 kDa
85.66	TFRC	Q90997	TFR1_CHICK	Transferrin receptor protein 1	84.0-86.0 kDa
95.36	EEF2	A0A1D5PS29	A0A1D5PS29_CHICK	Elongation factor 2	94.0-96.0 kDa
95.38	EEF2	Q90705	EF2_CHICK	Elongation factor 2	94.0-96.0 kDa
96.53	TFRC	A0A8V0YJE5	A0A8V0YJE5_CHICK	Transferrin receptor protein 1	96.0-98.0 kDa
112.23	ATP1A1	P09572	AT1A1_CHICK	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
116.32	ATP1A1	A0A8V1A740	A0A8V1A740_CHICK	Sodium/potassium-transporting ATPase subunit alpha	116.0-118.0 kDa
124.56	VCL	P12003	VINC_CHICK	Vinculin	124.0-126.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	10	0	0	Histone family	routine	, , AGT, GNRH2, [Histone family], , ADA2, , NTS, ME1
2.0-4.0 kDa	14	0	0		routine	NPPC, MLN, , SCT, AMPD1, GAL, RPN1, , HPX, , CLPS, SLN, BLACAT1, RPN2
4.0-6.0 kDa	34	0	0	Heat shock proteins; Mitochondrial markers	routine	C12orf75, , CRABP2, PYY, , , , , , TMSB4B, ,
6.0-8.0 kDa	142	0	0	Heat shock proteins; Mitochondrial markers	elevated	, , , PLN, PLN, TOMM5 [Mitochondrial markers], , , COP59, , C14orf2 [Mitochondrial markers], LOC107055098, , MT3, , TOMM7 [Mitochondrial markers], UQCRL1 [Mitochondrial markers]
8.0-10.0 kDa	213	0	0	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; p53 pathway	elevated	TMEM167A, , PKIA, , , , LOC100859314, GNG12, UBA52, ADM, , [Histone family], , , SERP1, GTF2H5, RPL38
10.0-12.0 kDa	272	0	1	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; p53 pathway	elevated	LOC418813, LOC121110032, SEC61B, , , , TCF4, , , , SRP9, , HIGD2A, SF3B5, , CXCL12,
12.0-14.0 kDa	314	1	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family	elevated	, LOC431325, SDHAF4 [Mitochondrial markers], , PTMA, , , LOC121107576, EMC6, , TMEM233, , , DNAL4, LBH, POLE4, ,
14.0-16.0 kDa	369	0	8	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	FAM103A1, H2A-IX [Histone family], IMMP2L [Mitochondrial markers], , , NDUFS6 [Mitochondrial markers], , , , , CCK, LOC121107574, UTS2, ISCA1 [Mitochondrial markers], TAC1, LSM10, TMEM35B
16.0-18.0 kDa	422	1	1	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; p53 pathway	elevated	, LITAF, , PNRC2, , , , UMAD1, MCFD2, LOC121108846, LIN52, GNG4, RPL22, CATHL1, PNRC2, LOC112531842, C19orf70, IL18BP
18.0-20.0 kDa	409	0	4	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family	elevated	MMGT1, UBE2J, GCSH [Mitochondrial markers], , SFT2D1, , CPLX4, PPP1R17, NGB, PLP2, , PPIL3, TCXET1D2, LOC107050964 [Mitochondrial markers], SAMD12, , NUDT1, UBALD1
20.0-22.0 kDa	456	0	9	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	, FAM180A, FAM168B, CRYAB, , , LOC121109670, TMEM47, MPLKIP [Actin family], LOC121108888 [Mitochondrial markers], , TRAPP3C, , TIMM17B [Mitochondrial markers], MRPS25, LOC768589, C5orf24, LOC100658647, ARF6
22.0-24.0 kDa	479	0	12	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	, VSTM2L, [Histone family], CLEC3A, TRNP1, DNAJC5B, COX5B [Mitochondrial markers], [Histone family], SNX24, C8FB, FAM219B, LOC121106526, LOC121108905, , BMF, LOC100858177, CD320, MRPS23
24.0-26.0 kDa	494	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	, TCXET1D4, PSMD10, CHMP2B, CDCDC59, TMEM140, RAB4A, XPMP4, FZD8, , TCEANC2, RPS8, THPO, CHMP2B, , LOC124416945, RAB19, VPS29L
26.0-28.0 kDa	479	2	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RABL3, TSN, , PPP1R2, COQ10B [Mitochondrial markers], RAB3C, , PRSS3, C1QC, , LOC107049069, TXNL4A, KLK7, LOC768817, SCAMP4, , EFNA5, GCH1
28.0-30.0 kDa	438	1	4	Actin family; Cell-cycle CDK/cyclin; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	RCHY1, KIAA1644, UTP23, FOLR3, RPL8, , ZCCHC9, , PEX11B, PLPP5, DOLPP1, LOC100859906, SRSF1, MVB12A, , CLTB, PSMA7, SCO2 [Mitochondrial markers]
30.0-32.0 kDa	451	1	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, ITM2B, TMEM220, DRAM2, HOXD12, TFAM [Mitochondrial markers], ELOVL3, SNAI2, ZFP511, AGPAT2, , LOC101747821, MYOZ2, PENK, UPRT, IGFBP5, TMEM106C, MRPL19
32.0-34.0 kDa	489	2	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, HUS1, , RSPQ3, AQP3, EMCN, NFU1 [Mitochondrial markers], ANKRD39, JAML, EXOSC8, KRT222, MMACHC [Lamin family], THEM4, LOC415758, NSL1, KCNM2B, MNR2, CENPQ
34.0-36.0 kDa	635	3	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UNG, MADPRT, SLC35E3, RASSF3, ORC6, , RAX, SPDEF, CD200R1A, REEP6, FAM168A, SIAH2, TMEM177, RX2, HOCX11, SPARC, SULT1B1, PP16
36.0-38.0 kDa	516	0	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	CHIR3B3, LOC112533538, MTERF4 [Mitochondrial markers], AASDHPTT, TMEM189, IRF1, PPP1R3C, TOB1, LOC121112382, APTX, NEIL2, ZFP36L1, STAR [Mitochondrial markers], LRRC23, TNFAIP1, , C14ORF166B, C1GALT1C1
38.0-40.0 kDa	505	0	16	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, NODAL, SNX20, RNF215, ABHD6, HSF2BP [Heat shock proteins], SORD, DUSP23, CD8A, LOC101751271, AHSA1 [Heat shock proteins], GALM, TNFRSF6B, EXTL2, BMERB1, HAGH [Mitochondrial markers], RNF146, TIMP2
40.0-42.0 kDa	495	2	13	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; p53 pathway	high	PARP11, SH3GL3, , CFAP410, IL10RB, FAM45A, SH3GL2, RPLP0, GNA72, WNT8C, NECAB3, PMEPA1, SETD7 [Histone family], GLT8D2, PEX16, RABEPK, TNFRSF9, ATP6V0D1
42.0-44.0 kDa	471	1	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	PRSS23, BGV, LOC100858953, ACTBL2 [Actin family], NR1I3, B3GNT3, ACTC1 [Actin family], TPST2, SGK494, ACTB [Actin family], HSD11B2, LYAR, B3GNT10, TPST2, LZTFL1, ACTA1 [Actin family], PGA5, GNA11
44.0-46.0 kDa	450	0	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers	high	MAPK9 [MAPK cascade], RPRD1B, SERPINE3, SEH1L, SERPINE2, OXTR, E2F1, LMX1B, LOC121109782, ONECUT2, RBM41, MAP2K2 [MAPK cascade], RPL3, SCEL, CCNA2 [Cell-cycle CDK/cyclin], CCDC94, MRPS35 [Mitochondrial markers], LOC107049146
46.0-48.0 kDa	471	1	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	NTSR1, LIMS1, GLA, SS18, SCRN1, ABRAXAS1, BMP4, CZH18ORF25, C17orf58, CCNA1 [Cell-cycle CDK/cyclin], AADAC3B, SPATA17, PKDCCB, NR2F1, KRT19, ALDH1A2, KDF1, PIP4K2A
48.0-50.0 kDa	426	1	14	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	high	KMT5A [Histone family], GLUL, TTL19 [Tubulin family], MTHFSO, KCTD18, BZW1, DRD3, SYT13, ABHD2, BMP6, TBX19, JMDJ6, LHX2, FFAR2L7, RCC1L, MHCY15, AP4M1, HS6ST1
50.0-52.0 kDa	429	2	6	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TUBB6 [Tubulin family], AZIN1, DLST [Mitochondrial markers], CHRNA10, MTERF2 [Mitochondrial markers], FAM186B2, NAF1, CHRNA10, CELF5, MFSD9, TUBA3E [Tubulin family], C4BPM, , NPY5R, PTAR1, C7orf26, CADM1, SERPINF1
52.0-54.0 kDa	378	0	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	high	PRR33, CRBN, CPQ, PTGER4, PPP2R2D, GLRA2, FAM134B, SUSDA4, F9, GFRA1, , HARB1I, PLD3, GPBP1L1, GDF10, TSEN2, NTD5C1, CARNMT1
54.0-56.0 kDa	422	0	11	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	TRMT13, IL17REL, HSPA13 [Heat shock proteins], SYT17, PLEKHD1, THUMPD2, HSPA14 [Heat shock proteins], MATN1, ENTPD8, SKA3, EVA1C, G3BP2, AP5M1, ATF2, MAPKAPK5 [MAPK cascade], CAMK2A, SLC2A1, HATR8L5
56.0-58.0 kDa	398	2	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	TMEM39B, SH2D2A, ALDH3B1, , LRRC71, DUS1L, CHRN82, ENTPD8, , EDC3, KATNA1, CHRNB4, PCYOX1, WASF3L, DFNAS, GABRB4, TMEM161B, CREB5
58.0-60.0 kDa	362	0	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	GJA10, LACTB, PKM, KPNA2, LIPI, RFX5, , SESN3, , ALDH7A1, IMPDH2, SNTG1, SLC68A1, SEPTINA4, CASP8 [Apoptosis caspase/BCL], CYP19A1, SLC16A8, MIER1
60.0-62.0 kDa	304	0	4	Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	YRK, ZFYVE28B, , SRC, ASIC1, ESAM, GPT2, CAT, DPY5, FZD5, PLD5, CHRM5, PSAP, CYP17A1, SLC2A2, SPATA2, ATP5F1AW [Mitochondrial markers], MSL1
62.0-64.0 kDa	278	0	3	Actin family; Heat shock proteins; Histone family; Mitochondrial markers	elevated	SPATS2L, PLBD2, GN3L, SLC43A2, DKC1, NEFL, SNTA1, MON1A, HAS3, KRT10, , NOTUM, THNSL2, FCRL4, , PFKFB3, CRISPLD2, ELP3
64.0-66.0 kDa	293	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	GPC5, IL2RB, UBQLN4, CNOT6, WFIIKN2, LOC107056275, GALNT1, HABP2, OPTN, A1CF, ASNS, PRKAA1, FANCC, SLAIN2, BPIL3, IL12RB1, , SLC27A2
66.0-68.0 kDa	247	2	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	RAP1GDS1, BFSP2, KLHL10, ENC1, PTCHD3, FBXW7, P2RX7, SLC2A1, CTPS2, GPR107, , CRY2, KLHL7, NEIL3, BTBD17, PLEKH02, AIFM3, KLHL21
68.0-70.0 kDa	249	1	6	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	RPA1, PTPN9, LOC419409, SLC18A2, KLHL20, COLGALT1, ANKRD13A, LOC121108224, GALNT9, ABCD4, PRKCI, SPATA7, LOC429032, PAFSS2, CERT1, CANX [ER markers], NTN1, SGO1
70.0-72.0 kDa	239	1	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	ENTPD1, ZBTB4A, CROT, LOC420419, RUFY2, TRAF4, ZBTB7C, ICK, PODXL2, XPNPEP1, NEURL1, SLCO2A1, ASIC4, EYA4, TNIP1 [Actin family], KLHL40, KRT3, KBTBD3
72.0-74.0 kDa	219	3	2	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	TMEM62, SLC6A14, HSPA5 [Heat shock proteins], MTMR6, FRMD1, RAD21, OTUD5, , PLPPR3, TRIM7.1, PIAS3, MED17, DDX3X, FBXL16, PiGR, , LOC121107539, FLAD1
74.0-76.0 kDa	175	0	3	AKT pathway; Actin family; Heat shock proteins; Histone family; Mitochondrial markers; NF-kB/Rel	elevated	TRIM47, DAGLB, BRD7, HEATR3, SCNN1G, SIMC1 [Actin family], SLCO1A2, SPTY2D1, KCNA4, PATZ1, RBM44, ACTR8 [Actin family], SLC25A13, ATXN1L, IL18RAP, PROS1, SENP1, KCNN3
76.0-78.0 kDa	198	0	8	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Mitochondrial markers	elevated	TBC1D23, HBSL1, LOC420374, RHPN1, POLH, NEK8, ZNFB3, LRRC45, TRIP4, ZBBX, ABCB10 [Mitochondrial markers], BFPSP1, KIFC1, CHFR, QSOX2, RACGAP1, SAMSN1, DYM
78.0-80.0 kDa	190	1	6	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Mitochondrial markers	elevated	TNFAIP2, LOC101750925, PARM12, FBXO34, DNAAF2, HSPA8 [Heat shock proteins], CHST4, GTSE1, SLC6A6, TBX3, PRMT7, CCDC110, PRMT7, SLC05A1, RBSN1, PNN, APBA3, SKIL
80.0-82.0 kDa	179	0	5	Actin family; Lamin family; MAPK cascade; Mitochondrial markers	elevated	SLC7A4, MTA1, , MRE11, SSRP1, SEMA3B, LEKR1, RASEF, SARMI1, SLC34A2, ABCG2, TERF2, HSD17B4, , TERF2, CIZ1 [Actin family], TMC03, LOC416530

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
82.0-84.0 kDa	195	3	7	Actin family; Heat shock proteins; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	CEP63, ANKRD13D, VRTN, SMYD4, USP49, SLC26A6, AK7, PAPOLG, XPR1, CEP63, IGSF21, ESPN, CNOT10, ACS2S, COLGALT2, POMT2, IMMT, EXT2
84.0-86.0 kDa	193	3	5	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	high	MTSS1L, SGIP1 [Actin family], NCKIPSD [Actin family], HSP90AA1 [Heat shock proteins], KIF2A, LRFN5, RUBCNL, SMURF1, EXOC8, KRIT1, HMHR, TSGA10, HSP90AA1 [Heat shock proteins], RPS6KA6, TRDN, FCHSD2, ZNF770, TMEM8A
86.0-88.0 kDa	186	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel	elevated	SIM1, MYBL1, MFN2, LOC107050560, MCMBP, C1R, PFKL, VPS51, SPG7, , ALOX5, APLP2, CHGB, GPHN, CYP1A2, HOMER3, CHUK [NF-kB/Rel], PAPOLB
88.0-90.0 kDa	178	0	6	Actin family; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	elevated	GCC1, LARGA1, PAMR1, RXFP1, THEMIS, PRDM4, HCFC2, , ZER1, RUSC1, SMURF2, , SATB1, , DACT2, STAT3 [JAK-STAT], RNF19A
90.0-92.0 kDa	139	0	4	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; MAPK cascade; Nuclear receptor PPAR	elevated	CSP2RA, HECTD2, HPS6, ECE1, LRRN4, ADAM9L, EZH1 [Histone family], DTNA, , SMO, TRPC1, GGNBP2, PIBF1, PRSS12, TTC14, PIK3AP1, AP521, PARP9
92.0-94.0 kDa	129	0	5	Actin family; Cell-cycle CDK/cyclin; Mitochondrial markers; Tubulin family	elevated	APBA1, SIDT1, DGKB, UHRF2, FGFR3, , , STRIP1 [Actin family], AFAP1 [Actin family], PROM2, DLGAP5, VPS35, , LOC421108, ARMC9, WDR27, , MYBL1
94.0-96.0 kDa	122	2	2	Actin family; Heat shock proteins; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	AGBL2, IKBKE [NF-kB/Rel], GPAM [Mitochondrial markers], SOBP, STOX1, BACH2, ZC3H12B, PRLR [Actin family], DZIP1, VAC14, POT1, DPP6, LOC428714, PRMT9, PRLR [Actin family], LRP3, CMIP, TRIM71
96.0-98.0 kDa	139	1	4	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Mitochondrial markers; p53 pathway	elevated	PLAG24EL6, , VWA2, POSTN, PRKD1, EXOC6, WDR75, CCSER1, TRPV4, ZXC, TMEM245, ADAMTS5, ARPP21 [p53 pathway], ATG9A, TRPV4, HELLS, TTC27, TTC27
98.0-100.0 kDa	105	0	7	Histone family; MAPK cascade; NF-kB/Rel; p53 pathway	elevated	ACO1, ADAM19, EXOSC10, CDH1, DLG3, PACS3, TGFBRAP1, LRP12, AGAP1, CLSTN2, PDE8B, MAP4K5 [MAPK cascade], PYGL, ESYT2, PPP6R3, MIOS, LOC422926, PIK3R5
100.0-102.0 kDa	111	0	2	Actin family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	MYCBPAP, PDE6C, SEZ6, CARNS1, PSMD2, GRM3, IL31RA, PTGFRN, CARNS1, TGS1, TBCK, MAP4K3 [MAPK cascade], GRIA3, DDIX10, INTS6, WFS1, CORO7, MTBP
102.0-104.0 kDa	111	0	3	Actin family; Apoptosis caspase/BCL; JAK-STAT; Mitochondrial markers	elevated	CEMPC, LOC101749378, DLG1, ANAPC2, NCOA7, KSR1, CHD1L, RBM15, COL5A2, HKDC1, SLC4A1, ZPAX, ABCB6, ZZZ3, ARHGAP12, ZHX3, PCDHGC5, MGEA5
104.0-106.0 kDa	112	0	4	Actin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	ECT2, SLC26A8, N4BP1, EDEM3, NBR1, TLR15, ACTN4 [Actin family], ACTN4 [Actin family], MAP3K14 [MAPK cascade], INTU, ACTN2 [Actin family], AP2A2, USP43, ODF2, EXOC1, GRIK1, CMIP, RNF216
106.0-108.0 kDa	97	0	4	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; NF-kB/Rel	elevated	PHF14, CWF19L2, MDGA1, PSMD1, FAM193B, TRIM24, DNAJC6, SETD6, LOC107049856, ANO6, BCLAF1 [Apoptosis caspase/BCL], AMER3, UBA7, HACE1, DGKG, CTNND1, MIB2, NFKB2 [NF-kB/Rel]
108.0-110.0 kDa	99	0	3	Histone family; JAK-STAT; MAPK cascade; NF-kB/Rel; Tubulin family	elevated	NEK10, EFTUD2, PCDH8, NCOA7, NFKB1 [NF-kB/Rel], BICC1, GRM7, OSBPL1A, TUBGCP2 [Tubulin family], CC2D1B, LOC101748987, SH3BP4, TMC1, PLXNC1, DCLRE1A, HDAC7 [Histone family], GRIK4, PAM
110.0-112.0 kDa	90	0	4	Actin family; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; Mitochondrial markers	elevated	MMS19, PHF12, SRBD1, ELAPOR2, MIB1, KIF5B, TDP1, DSE, EMCL, RTN4, KANK3, PKM2, STAT6 [JAK-STAT], ROBO4, DGKI, SCUBE3, BNC1, SCARF1
112.0-114.0 kDa	80	1	1	Actin family; Histone family; Mitochondrial markers; Tubulin family	elevated	IARS2 [Mitochondrial markers], LOC112530482, ATP1A2, INPP4A, KCNH5, HIRA, KANSL1L, USP15, DPYD, EPHB2, WDR59, ATP1A1, , EPHA5, TMC07, CCDC146, NAA25, HYOU1
114.0-116.0 kDa	86	0	2	Actin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	UBAP2L, ZCCHC2, SAMD11, ULK1, HK3, HPS3, PCDH10, RIN2, TMPPRS15, SNX13, TUBGCP3 [Tubulin family], CNTN2 [Actin family], PDE10A, CBLB, POP1, AASDH, TAF4, ZSWIM5
116.0-118.0 kDa	71	1	0	Actin family; MAPK cascade; Mitochondrial markers	elevated	PLEKHG3, ITGA8, CLB, RECQL5, ITGA8, FBXO18, MAN2C1, SEMA6A, KANSL3, ANKRD28, ATP1A1, CLIP2, FUK, ZFPM1, PTK7, COL15A1, PRPF4B, HERC3
118.0-120.0 kDa	58	0	0	Actin family; Histone family	routine	CASR, OFCC1, IDE, MORC1, UBAP2L2, PCDH19, GRIP2 [Actin family], LRRCC1, INTS3, TOP3A, CD44, WDR60, SREBF1, ZBTB11, TSHZ1, LLLG2, USP48, ICE2
120.0-122.0 kDa	73	0	2	Actin family; Histone family; Lamin family; Tubulin family	elevated	DIS3L, SEC24A, IPO7, HIP1R [Actin family], SMARCA1, SSH1, B4GALNT3, GRID2IP [Actin family], LOC107048990, STK11IP [Actin family], STK11IP [Actin family], EHPB1L1, PK3CD, SSGM2, AGBL1 [Tubulin family], NLRC3, NCAM1, SREBF2
122.0-124.0 kDa	86	0	4	Actin family; Histone family; Nuclear receptor PPAR; Tubulin family	elevated	PIK3CB, PTK7, SMARCC1, SRGAP1, LRIG3, KDM4C [Histone family], LRRFIP1 [Actin family], MIS18BP1, BUB1, PLD1, AP3B1, SHD19, , ALS2, TMEM132D, ZNF3, ATOSA, NPR2
124.0-126.0 kDa	62	1	2	Actin family; JAK-STAT; p53 pathway	elevated	XPO7, SRGAP3, UBE3C, GMIP [Actin family], MYO1H, BMP2K, ARHGAP45, SLC38A10, MEGF10, PIK3CA, KCNH8, SEC24C, RC3H1, ELMSAN1, ADGRD2, ZNF654, GRIN3A, SLC4A10
126.0-128.0 kDa	68	0	0	Actin family; Lamin family	routine	MYO1C, CORIN, CKAP2L, ADAR, SIN3B, FARP1, CEP131, LAMB3 [Lamin family], MKL1, DENND2B, SLC12A2, KIAA1522, MYRF, , LAT52, TEK, PPP2R3A, POLR1B
128.0-130.0 kDa	63	0	3	Actin family; JAK-STAT; Lamin family; Mitochondrial markers; p53 pathway	elevated	, PIK3CG, SLC4A5, ZFPM2, SMC5, TOPORS, MICAL1, USP7, FNIP1 [Actin family], , ADCY1, ERC1 [Actin family], COBLL1, USP7, JARID2, SLC12A4, PDZD8, PTK2
130.0-132.0 kDa	47	0	1	Actin family; Apoptosis caspase/BCL	routine	PER3, CEP128, HEATR6, TMEM132A, DAB2IP [Actin family], BLM, XPO4, ABL2, VWA3A, TRERF1, USP25, DUSP27 [Actin family], FAM129A, ITGA7, LOC101750972, ITGA11, PRTG, PRTG
132.0-134.0 kDa	62	0	0	Actin family	routine	GPRI58, PKP4, TIE1, MYO1B, CWF19L1, MUC6, FBLN2, TRPA1, TBC1D8, ATP11B, ZBED4, ATP11C, NRDC, FMNL1, SEC33B, ZNF516, , ADCY6
134.0-136.0 kDa	69	0	4	Actin family; JAK-STAT	elevated	NCKAP1L, , CGN, TAF2, WASHC5, ATP8B4, ATP13A4, ZMYND8, CAPRIN2, PRDM15, ATP13A4, RAB11FIP1 [Actin family], MOV10L1, MYO1E, TRPM5, AFF3, SMARCC2, JAK1 [JAK-STAT]
136.0-138.0 kDa	48	0	0		routine	, WASHC4, ADAMTS3, PCDH7, CFAP61, REXO1, ZNF687, TNKS, HK2, CAND1, RAPGEF1, LOC121108097, GDF2, NID1, , B4GALNT4, KCNH6,
138.0-140.0 kDa	46	0	0	Actin family; Lamin family; Tubulin family	routine	NOL6, GRM5, RAPH1, JARID2, EGF, REV1, CFAP70, EDRF1, SUCO, LARS, TBDC [Tubulin family], COPA, AUTS2, BRPF1, DGKH, , PHKA1, ATP13A3
140.0-142.0 kDa	48	0	3	Actin family; Apoptosis caspase/BCL; Histone family; MAPK cascade; NF-kB/Rel	elevated	NEK1, KCNM1A, TJP2, BRIP1, PPFIA4 [Actin family], MAP3K6 [MAPK cascade], ADGRA2, MECOM [Histone family], ATP11A, ATP2B2, ADCY8, ATP13A1, MAMDC4, NRDE2, AH1, DIAPH2, IGF1R, EHPB1
142.0-144.0 kDa	44	0	2	Actin family; MAPK cascade	routine	ATN1, DCTN1 [Actin family], MMS22L, ZMYM2, HDLBP, MYBPC3, SPAG5, TRPS1, PLCB2, , PRAG1, MCF2L, GRIN2C, CTC1, FBRSL1, AMBRA1, CSPP1, SF1
144.0-146.0 kDa	49	0	0	Actin family; Histone family	routine	RRP12, STAG1, ZNF592, TBC1D4, MATN2, ZNF804A, MTUS1, USP40, ABCC13, ABCB1LA, PLEKHA5, SCAP, ALPK3, STAG2, FLJ1, USP31, SMC1B, MYBPC3
146.0-148.0 kDa	36	0	1	Actin family; Histone family; Tubulin family	routine	, VEGFRKDRL, PPFIA2 [Actin family], STK36, WDR17, EHMT1 [Histone family], CENP1, SORBS1, INF2, AOX1, IARS, ERB4, TTBK2, ADGRA3, ACE, ARHGEF9, MYPN, SEC31A
148.0-150.0 kDa	49	0	2	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; MAPK cascade	elevated	PER2, CNTNAP1 [Actin family], NEDD4, CADPS, PLCG2, RNF123, ZNF521, KANK1, RPGR, FAM21C, RTE1L, TNR, POG2, DISP3, PROB1, NEURL4, GAK [Cell-cycle CDK/cyclin], LOC415478
150.0-152.0 kDa	42	0	0	Actin family; Histone family	routine	SIK3, NRXN1, KDR, PDSSA, BLM, ADCY9, IKBKAP, LOC100858919, PHLDB2, , TTBK1, UPF2, FNDC1, XDH, CCDC18, CPD, LOC107050768 [Histone family], IQSEC1
152.0-154.0 kDa	39	0	2	Actin family; Histone family; MAPK cascade	routine	MAP3K1 [MAPK cascade], ABCC4, UMODL1, MAP3K5 [MAPK cascade], NFASC, PNPLA7, TCP10, SH3TC1, FHOD1, INSR, VWA3B, ADGRG2, SYNRG, PHLPP2, SOS2, COL17A1, COB1,
154.0-156.0 kDa	27	0	0	Actin family	routine	NUP155, PTPRJ, PTPRJ, TRIOBP [Actin family], EDCA, CRB1, MET, DHX57, MYO3B, ANKRD50, ZFYVE9, DROSHA, IGSF9, DHX29, BRD4, PALD, FLT1, RPRD2
156.0-158.0 kDa	30	0	0	Actin family; Histone family	routine	ATRN [Actin family], ATAD2, NHP4, LOC424998, CUX2, LCOR, FLT4, TERT, ARHGEF10L, IMPG2, , IGSF9B, POLR3A, USP47, PITPNM2, USP47, HMGXB3, GPATCH8
158.0-160.0 kDa	19	0	1	Actin family	routine	PCMTD2, , STARD13, MED23, NCOA2, RAB11FIP5 [Actin family], SCAF11, ROCK1, MAP4K4, FAM135B, NEO1, NISCH, AIM1L, RIC1, ATP10D, CEP164, PTPRG, ZMYM3
160.0-162.0 kDa	42	0	0	Cell-cycle CDK/cyclin; Histone family	routine	KDM6A [Histone family], MED14, SYNGAP1, CRB2, KIF7, ZNF423, MRC1, USP19, DAPK1, PTCH1, LRPPRC, COL20A1, FGD6, , PTCH1, MEGF6, FANCD2, COL4A1
162.0-164.0 kDa	28	0	1	Actin family; Cell-cycle CDK/cyclin	routine	DISP2, PTPRK, A2ML2, LTK, PLEKH11, COL4A3, FAM193A, MYOM3, CEP162, CDK12 [Cell-cycle CDK/cyclin], KTN1, ZNF608, ASXL2, KIF15, ATAD2B, UHRF1BP1, CASKIN1 [Actin family], PPIPF5K1
164.0-166.0 kDa	38	0	1	Actin family; Mitochondrial markers	routine	SETD5, KIF24, CPS1 [Mitochondrial markers], MYOM2, IRAG2, LRRC7, COL4A4, SORCS1, WDR7, A2ML4, CECR2, PDSSB, CLASP1, COL4A6, CASKIN2, NHS [Actin family], RGS12, PIK3R4
166.0-168.0 kDa	32	0	3	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Nuclear receptor PPAR	routine	TRAPPC8, SRCIN1, OYST, EEA1, , KIAA2022, TEX14, GLI2, ADGRB2, NKTR, SNAPC4, MAGI2, RALGAPB, ARHGAP31, EIF3A, CDK13 [Cell-cycle CDK/cyclin], ZMYMA, ADGRL2
168.0-170.0 kDa	22	0	0		routine	BCORL1, CFTR, GRIN2B, ADGRL3, PLA2R1, SLIT1, CAMSAP2, COL4A2, TOPBP1, AB13BP, TA3, RERE, PHLPP1, FAM179B, OVSTL, ERCC6, CLASP2, ABCC6
170.0-172.0 kDa	21	0	2	AKT pathway; Actin family	routine	PIK3C2G, CRACD [Actin family], PTPRS, DIP2B [Actin family], FANCA, ABCU3, POLA1, GOLGA3, TRPM2, DLC1, ADGRB3, GLI3, BAZ1B [AKT pathway], ATP10A, LMTK2, ANKRD31, CLUH, ABCC3
172.0-174.0 kDa	14	0	0		routine	SLIT3, ARHGAP5, COL24A1, ZNF518A, ABCC1, PLEKHG1, A2ML3, DNMT1, KDM5B, ABCC2, SOGA1, ALK, LOC121109116, AHDG1
174.0-176.0 kDa	20	0	0		routine	TARBP1, NRXN3, TNS3, TULP4, AQR, AGL, UGGT1, TTC37, DNMBP, PLEKH2, TOP2A, FAM135A, TOP2A, RIMS2, DUOX1, SYNJ1, ARHGEF18, ABCC9
176.0-178.0 kDa	24	0	0	Actin family; Histone family; Lamin family	routine	HECW2, , KDM5B [Histone family], RUSC2, LAMC1 [Lamin family], BAZ2A, COL16A1, DIP2A [Actin family], RIMS1, DIP2C [Actin family], COL27A1, CCDC180, ZCCHC6, LRR1Q1, , ADAMTS12, CC2D2A, TET3
178.0-180.0 kDa	17	0	0	Actin family; Lamin family	routine	CTTNBP2 [Actin family], SMARCA2 [Actin family], RREB1, SORL1, TIAM1, PAPPA2, , UGGT2, DNMT1, [Lamin family], TOP2B, FYCO1, DISP1, CCDC171, LAMC3 [Lamin family], COL18A1, IQGAP3
180.0-182.0 kDa	19	0	1	MAPK cascade	routine	PLCH2, LRP6, PXDN, URB2, WDR62, ABCA9, , LTBP1, LRP5, DEPDC5, RIMBP2, , MAP3K4 [MAPK cascade], PAPPA, PARP4, TNKS1BP1, ROBO1, CAMSAP1
182.0-184.0 kDa	19	0	2		routine	COL11A1, ALS2, HEG1, SYNM, EIF4G1, SAMD9L, ADGB, THOC2, ERCC6L2, RB1CC1, EIF2AK4, KIF21B, CEP170B, LOC101749352, TOP2B, , SPHKAP, C4
184.0-186.0 kDa	19	0	0	MAPK cascade	routine	MYO15L, KIF14, USP6, CAMTA1, ABCA8, FHOD3, RAPGEF2, DLEC1, HECW1, LOC100858386 [MAPK cascade], ZCCHC11, TRPM1, CROCC, MADD, PBRM1, , SHANK3, RAPGEF6

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
186.0-188.0 kDa	15	0	0		routine	, MBD5, ABCA5, PCF11 , , THSD7A, PREX1, PBRM1, DNAAFB, TNS1, LOC101748706, KIF21A, VTG3, C5, NHSL1
188.0-190.0 kDa	15	0	0	Histone family	routine	PRDM2, FRMPD3, MARF1, PHRF1, DOT1L [Histone family], COL5A1, EPRS, , RREB1, LTBP2, IQGAP1, COL1A4, MON2, KMT2E, C4A
190.0-192.0 kDa	10	0	1		routine	ITSN1, TMEM131L, PLCH1, IQGAP2, SIPA1L2, SLX4, EIF4G3, MRC2, BRCA1, PIK3C2A
192.0-194.0 kDa	19	0	0	Actin family	routine	RICTOR, TIAM2, CLTC, KDM5A, BCOR, FAM208A, TRPM3, NES, ESPL1, CFAP74, MARF1, FMN2, PREX2, POLR1A, MICAL2 [Actin family], ARAP2, ZC3H13, SI
194.0-196.0 kDa	17	0	0	Lamin family	routine	, , CFAP43, KDM3B, FRMPD4, ARAP3, ABCA3, ARHGEF28, SHPRH, USP54, CLTCL1, CDC42BP4, LAMB2 [Lamin family], , CDC42BPB, MYO5B, CDCC141
196.0-198.0 kDa	15	0	0	Lamin family	routine	SHROOM2, LAMB1 [Lamin family], PEAK1, NUP98, MYOM1 , , PSME4, CHD1, ARID2, SIPA1L1, MGAM, CFAP65, AIM1, RAB44, CFAP65
198.0-200.0 kDa	16	0	0	Actin family; Lamin family; Tubulin family	routine	NUP188, KIDINS220 [Actin family], TNC, NRAP, QSER1, GCC2, ZSWIM8, LOC101748153 [Lamin family], N4BP2, LTN1, LY75, MCM3AP, LTN1, TUBGCP6 [Tubulin family], ADAMTSL3, UBR2
200.0-202.0 kDa	22	0	0	p53 pathway	routine	EPB41L1, NALCN, ADAMTS7, ADAMTSL1, UNC13A, FOCAD, NRCAM, KIF1A, VWDE, UBR1, ALPK2, SUPT6H, LOC112531056, ARFGEF2, XRN1, BAZ2A, WDR90, BRWD3
202.0-204.0 kDa	15	0	0	Lamin family	routine	ARHGEF5, LAMB4 [Lamin family], LOC416924, KIAA1671, LAMA4 [Lamin family], COL14A1, IFT172, MYO5C, VIRMA, COL14A1, SCN11A, TMEM131, LOC426820, CEP152,
204.0-206.0 kDa	15	0	1	p53 pathway	routine	GSE1, ECPAS, TSC2, KIF1B, LARP4B, VTG2, ITSN2, CHD1W , , SHANK2, TP53BP1 [p53 pathway], KIAA1549 , , ATAD5, DOCK2B
206.0-208.0 kDa	12	0	0	Actin family	routine	BTAF1 , , VTG2, PHIP [Actin family], LOC424300, CPAMD8, RAI1, PATJ, ZNF106, AK9, SCN4A, NLRCS
208.0-210.0 kDa	11	0	0		routine	PCSK5, SCRIB, ITGB4, CHD1, TNRC6A, LOC101747860, PPL, Pcdh15, PLXNB2, PDCD11, ARFGEF1
210.0-212.0 kDa	18	0	0	Histone family; p53 pathway	routine	PLXNA2, LOC421255, LMO7, Mylk, RESF1, VTG1, VTG1, UBR3, TANC1, FER1L6, DENND4A, TCF20, ZNFX1 , , KAT6A [Histone family], GBF1, NUP210 [p53 pathway], TAF1
212.0-214.0 kDa	15	0	0	Actin family; p53 pathway	routine	KIAA1217, ZNF236, NUP214 [p53 pathway], MYO5A, PLXNA4, MAP2, PCM1, CEP295, CHD2, TJP1, PTPRF, LRP4, VWA8, TICRR [Actin family], CACNA1S
214.0-216.0 kDa	17	0	0		routine	SBF2, SCN8A, PTPRD, MAST2, ADAMTS20, DOCK2, PLXND1, ANK1, DOCK1, AFDN, WDR81, DDX60, THADA, CFAP44, MTCL1, MYO16, PLXNA1
216.0-218.0 kDa	12	0	0	Actin family	routine	RLF, Xirp1 [Actin family], THADA, ANAPC1, KIF16B, KIF20B, DENND4C, KNDC1, AKAP11, NCOA6, KIF13B, AKAP12
218.0-220.0 kDa	15	0	0		routine	LCT, CHD4, KIF26A, DICER1, DICER1, SBF1, F5, EXPH5, PTPRS, CHD5, PRG4, EML6, NAV1, TANC2, WIZ
220.0-222.0 kDa	8	0	0		routine	GREB1L, HELZ, TET2, RALGAPA2, CACNA1F, GREB1, COL6A6, MIA3
222.0-224.0 kDa	22	0	1	Histone family; p53 pathway	routine	ADAMTS9, ARHGAP21 [p53 pathway], DOCK5, KIF13A, MYH1B, MYO7L2, SETD1B [Histone family], RBBP6 , , MYH1F, KNL1, MYH1C, ZNF142, LRRK1 , , SHROOM3, ACAN, ACAN
224.0-226.0 kDa	14	0	1	Apoptosis caspase/BCL	routine	FANCM, EML5, MYH15, MYLK, HEATR5A, HEATR5B , , MYH7B, AGRN, DSCAM, CASP8AP2 [Apoptosis caspase/BCL], MPDZ, TDRD15, ZFC3H1
226.0-228.0 kDa	10	0	0	Histone family	routine	KAT6B [Histone family], TTC3, LOC395944, CKAP5, MYH9, MYH9, KIF26B, SCN9A, TRIP11, PCM1
228.0-230.0 kDa	16	0	0		routine	MYH11, SCN3A, SCN2A, NUP205, ARHGAP33, SCN1A, TNC, NCKAP5, MYH11, KIAA2026, ANKRD12, SMCHD1, PLEKHG4, ZGRF1, OTOF,
230.0-232.0 kDa	6	0	0		routine	DLG5, DOCK4, SETD1A, PI4KA, CCDC88B, MYH7
232.0-234.0 kDa	10	0	0		routine	SCN5A, PHF3, MYH10, LOC420486, AGRN, TSPOAP1, ATG2A, OVOA, MUC5B, TECTA
234.0-236.0 kDa	4	0	0	Actin family	routine	MYOF, SPEF2, LOC421441, CIT [Actin family]
236.0-238.0 kDa	11	0	0		routine	BAZ2B, MYO10L, PCNX2, CR1, DOCK11, TECTA, MYO9B, FER1L4, DOCK3, SYTL2, MYO10
238.0-240.0 kDa	10	0	1	Cell-cycle CDK/cyclin	routine	MALRD1, CCDC88A , , DOCK8, CDK5RAP2 [Cell-cycle CDK/cyclin], NIN, SDK2, BAZ2B, SDK1, LOC430303
240.0-242.0 kDa	10	0	0		routine	ABCA7, MED13, SDK2, USF3, PIKFYVE, DOCK9, BDP1, SNRNP200, KIAA1549L, MED12L
242.0-244.0 kDa	9	0	0		routine	ASXL3, DOCK7, CACNA1C, DYSF, MED12, ARID1A, FRMPD2, GTF3C1, MED13L
244.0-246.0 kDa	5	0	0	Actin family	routine	PDE4DIP [Actin family], CABIN1, ARID1B, GON4L, INTS1
246.0-248.0 kDa	11	0	0		routine	SORL1, FREM1, ARFGEF3, CACNA1I, CLIP1, PRRC2B, OBSL1, PRR14L, EVPL, PTPRB, MYO7B
248.0-250.0 kDa	8	0	0		routine	WNK2, PLXNB3, YLPM1, TET1, CACNA1D, RTTN, SPAG17, ZNF318
250.0-252.0 kDa	3	0	0		routine	SDK1, PLCE1, ascc3
252.0-254.0 kDa	6	0	0		routine	CCDC162, NAV3, LOC415414, HEATR1, ZNF407, AHCTF1
254.0-256.0 kDa	8	0	0	Actin family; Heat shock proteins	routine	ABCA4, MICAL3 [Actin family], KIAA0100, RIF1, RIF1, ABCA1, DNAJC13 [Heat shock proteins], KNTC1
256.0-258.0 kDa	4	0	0		routine	C2CD3, CAD, CENPE, DOP1B
258.0-260.0 kDa	5	0	0		routine	CSPG4, PKDREJ, PTPRZ1, MYO7A, SVIL
260.0-262.0 kDa	8	0	0		routine	APC2, POLE, ROS1, ROS1, BRWD1, IGFN1, LOC416696, CUL9
262.0-264.0 kDa	9	0	0		routine	PTPRVP, LOXHD1, NOTCH2, SPTB, ACAC, CACNA1G, SPTBN2, NUMA1, LOC421195
264.0-266.0 kDa	5	0	0	Actin family	routine	NAV2, AKAP6, MPRIP [Actin family], ACACA, CACNA1B
266.0-268.0 kDa	5	0	0		routine	GOLGA4, SEC16A, CNOT1, ARHGEF4, ACACB
268.0-270.0 kDa	7	0	0	Histone family	routine	URB1, EP300 [Histone family], HIVEP3, IGSF10, CREBBP [Histone family], NBAS, CNTRL
270.0-272.0 kDa	6	0	0	Lamin family	routine	TTC28, CSPG4B [Lamin family], PCNX1, TLN2, TLN1, TLN1
272.0-274.0 kDa	7	0	0		routine	PKD1L2, ANKRD17, PDZD2, CACNA1E, FN1, PRPF8, SPG11
274.0-276.0 kDa	6	0	0		routine	SPTBN1, NCOR2, FASN, DIDO1, ATRX, CACNA1H
276.0-278.0 kDa	4	0	0		routine	NOTCH1, TPR, CEP192, STAB2
278.0-280.0 kDa	6	0	0		routine	FAM208B, PTPN13, GVINP1, SON, HIVEP2, JMJD1C
280.0-282.0 kDa	7	0	0		routine	POLQ, DOPEY2, CADNL, ABCA2, FASN, IGF2R, FBN3
282.0-284.0 kDa	6	0	0	Lamin family	routine	MAP1B, MKI67, MAST4, NCOR1, STAB1, FLNB [Lamin family]
284.0-286.0 kDa	8	0	0	Actin family	routine	ZNF462, EIF4BP3, PIEZO1, LRRK2, SPTAN1, ZNF292, XIRP1 [Actin family], ANKRD26
286.0-288.0 kDa	4	0	1		routine	HECTD1, CRYBG3, MTOR, SPTAN1
288.0-290.0 kDa	2	0	0		routine	CEP290, AKAP13
290.0-292.0 kDa	5	0	0	Histone family	routine	USP9, ZFYVE26, LOC107050955, CELSR2, SETD2 [Histone family]
292.0-294.0 kDa	4	0	0		routine	EPG5, PRRC2C, GCN1, USP24
294.0-296.0 kDa	1	0	0		routine	COL7A1
296.0-298.0 kDa	3	0	0		routine	LOC417131, RALGAPA1, HIVEP1
298.0-300.0 kDa	3	0	0		routine	LOC121110182 , , OTOGL
300.0-302.0 kDa	3	0	0		routine	FBN2, LOC427025, ATR
302.0-304.0 kDa	4	0	0		routine	TENM1, MYO9A, WNK1, TENM1
304.0-306.0 kDa	3	0	0		routine	CEP250, ITPR3, UNC79
306.0-308.0 kDa	5	0	0		routine	CHD6, NBEAL1, ITPR2, TENM2, TG
308.0-310.0 kDa	4	0	0		routine	ANKRD11, TENM4, NBEAL2, TENM3
310.0-312.0 kDa	6	0	0	Histone family	routine	UBR5, NSD1 [Histone family], LOC414835, TENM2, MUC5ACL, FBN1
312.0-314.0 kDa	3	0	0		routine	VWF, APC, SETX
314.0-316.0 kDa	1	0	0		routine	NIPBL
316.0-318.0 kDa	5	0	0		routine	ITPR1, TNRC18, ALMS1, RP1,
318.0-320.0 kDa	1	0	0		routine	UTP20
320.0-322.0 kDa	2	0	0		routine	NF1, HELZ2
322.0-324.0 kDa	4	0	0		routine	PIEZO2, LOC429249, LRBA, MXRA5
324.0-326.0 kDa	2	0	0		routine	CHD9, TACC2
328.0-330.0 kDa	2	0	0		routine	NBEA, BOD1L1
330.0-332.0 kDa	1	0	0		routine	DSP
334.0-336.0 kDa	1	0	0		routine	ZZEF1
336.0-338.0 kDa	2	0	0	Histone family	routine	ASH1L [Histone family], EP400

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
338.0-340.0 kDa	5	0	0		routine	CHD7, KALRN, LOC776067, COL6A3, FRYL
340.0-342.0 kDa	5	0	0	Lamin family	routine	COL12A1, LAMA1 [Lamin family], COL12A1, BPTF, CENPF
342.0-344.0 kDa	3	0	0		routine	DMXL2, DMXL1, HTT
344.0-346.0 kDa	1	0	0		routine	TRANK1
346.0-348.0 kDa	1	0	0		routine	FRY
348.0-350.0 kDa	3	0	0		routine	COL6A3, MGA, ATM
350.0-352.0 kDa	7	0	0	Lamin family	routine	, TRIO, LAMA3 [Lamin family], CELSR3, FREM2, CPLANE1, DCHS1
354.0-356.0 kDa	2	0	0		routine	C1HXORF59, CEP350
356.0-358.0 kDa	3	0	0		routine	REV3L, MAP1A, WDFY4
358.0-360.0 kDa	1	0	0		routine	CFAP54
360.0-362.0 kDa	1	0	1	NF-kB/Rel	routine	RELN [NF-kB/Rel]
362.0-364.0 kDa	2	0	0		routine	VPS13A, GOLGB1
366.0-368.0 kDa	1	0	0		routine	
368.0-370.0 kDa	2	0	0	Lamin family	routine	LAMA2 [Lamin family], CDH23
370.0-372.0 kDa	1	0	0		routine	LOC121109863
372.0-374.0 kDa	3	0	0		routine	UNC80, VCAN, SPEG
374.0-376.0 kDa	1	0	0		routine	LOC101751678
376.0-378.0 kDa	1	0	0		routine	BRCA2
378.0-380.0 kDa	1	0	0		routine	PTPRQ
380.0-382.0 kDa	1	0	0		routine	OTOG
388.0-390.0 kDa	3	0	1	NF-kB/Rel	routine	VCAN, RELN [NF-kB/Rel], CSMD1
392.0-394.0 kDa	1	0	0		routine	CSMD2
394.0-396.0 kDa	2	0	0		routine	ZFHX4, WDFY3
396.0-398.0 kDa	1	0	0		routine	ZFHX4
398.0-400.0 kDa	2	0	0		routine	ASPM, UTRN
400.0-402.0 kDa	3	0	0		routine	ZFHX3, CUBN, SVEP1
402.0-404.0 kDa	1	0	0		routine	
404.0-406.0 kDa	3	0	0	Lamin family	routine	USP34, CSMD3, LAMA5 [Lamin family]
406.0-408.0 kDa	1	0	0		routine	BSN
410.0-412.0 kDa	2	0	0		routine	SMG1,
412.0-414.0 kDa	2	0	0	Histone family	routine	SZT2, KMT2A [Histone family]
422.0-424.0 kDa	3	0	0		routine	VPS13C, DMD, ANK2
426.0-428.0 kDa	2	0	0		routine	LYST, DMD
436.0-438.0 kDa	1	0	0		routine	TRRAP
438.0-440.0 kDa	1	0	0		routine	ZNF469
440.0-442.0 kDa	2	0	0		routine	PKHD1, FRAS1
442.0-444.0 kDa	1	0	0	Heat shock proteins	routine	HSPG2 [Heat shock proteins]
448.0-450.0 kDa	1	0	0		routine	VPS13B
452.0-454.0 kDa	3	0	0		routine	ABCA12, SPTBN5, DNAH12
456.0-458.0 kDa	1	0	0		routine	AKAP9
458.0-460.0 kDa	1	0	0		routine	DNAH7
460.0-462.0 kDa	1	0	0		routine	ABCA13
462.0-464.0 kDa	1	0	0		routine	DNAH3
464.0-466.0 kDa	1	0	0		routine	FAT4
472.0-474.0 kDa	1	0	1		routine	PRKDC
474.0-476.0 kDa	3	0	0		routine	HECTD4, LRP1B, ANK3
480.0-482.0 kDa	1	0	0		routine	DNAH1
484.0-486.0 kDa	1	0	0	Actin family	routine	PKD1 [Actin family]
486.0-488.0 kDa	1	0	0		routine	DYNC2H1
488.0-490.0 kDa	1	0	1		routine	PRKDC
490.0-492.0 kDa	1	0	0		routine	VPS13D
494.0-496.0 kDa	1	0	0		routine	FAT2
498.0-500.0 kDa	2	0	0		routine	CMYA5, DNAH9
504.0-506.0 kDa	1	0	0		routine	PCNT
506.0-508.0 kDa	4	0	0		routine	DNAH17, BIRC6, LRP1, LRP1
508.0-510.0 kDa	2	0	0		routine	FAT3, FAT1
518.0-520.0 kDa	1	0	0		routine	MYCBP2
522.0-524.0 kDa	3	0	0		routine	SACS, APOB, LRP2
526.0-528.0 kDa	2	0	0		routine	DNAH10, LOC101748756
528.0-530.0 kDa	1	0	0		routine	HERC2
532.0-534.0 kDa	2	0	0		routine	DNAHSL, DYNC1H1
534.0-536.0 kDa	2	0	0		routine	DNAH5, HERC1
536.0-538.0 kDa	1	0	0		routine	DNAH8
546.0-548.0 kDa	1	0	0	Histone family	routine	KMT2C [Histone family]
552.0-554.0 kDa	1	0	0		routine	RYR3
556.0-558.0 kDa	1	0	0		routine	USH2A
558.0-560.0 kDa	4	0	0		routine	BLTP1, LOC415641, STARD9, SSPO
560.0-562.0 kDa	2	0	0		routine	SSPO, PCLO
564.0-566.0 kDa	1	0	0		routine	RYR2
566.0-568.0 kDa	1	0	0		routine	HYDIN
568.0-570.0 kDa	1	0	0		routine	PCLO
572.0-574.0 kDa	2	0	0	Histone family	routine	KMT2D [Histone family], UBR4
590.0-592.0 kDa	1	0	0		routine	RNF213
604.0-606.0 kDa	1	0	0		routine	HMCN1
632.0-634.0 kDa	1	0	0		routine	MDN1
684.0-686.0 kDa	1	0	0		routine	ADGRV1
700.0-702.0 kDa	1	0	0		routine	AHNAK2
736.0-738.0 kDa	1	0	0		routine	NEB
804.0-806.0 kDa	1	0	0		routine	SYNE2
830.0-832.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
882.0-884.0 kDa	1	0	0		routine	DST
914.0-916.0 kDa	1	0	0		routine	OBSCN

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
3894.0-3896.0 kDa	1	0	0		routine	

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.64		P83308	FMRFamide-like neuropeptide	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.13		P19851	Neurokinin-A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.23	AGT	P67885	Angiotensinogen (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.25	GNRH2	P68072	Gonadoliberin-2	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27		P84553	Histone H1A2 (Fragment)	0.0-2.0 kDa	Histone family	configured audit family match: Histone family
1.38		P19850	Substance P	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.47	ADA2	P84520	Adenosine deaminase 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.55		A0ABV0XA68	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.61	NTS	P13724	Neurotensin	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.84	ME1	Q92060	NADP-dependent malic enzyme (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.24	NPPC	P21805	C-type natriuretic peptide	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.69	MLN	Q9PRP6	Motilin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.97		A0ABV1A5Z4	LASP1 neighbor	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.13	SCT	P01280	Secretin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.19	AMPD1	P81073	AMP deaminase 1 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.21	GAL	P30802	Galanin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.24	RPN1	P80896	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.27		A0ABV1ADG6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.38	HPX	P20057	Hemopexin (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.39		P20137	Glutathione S-transferase 5 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.48	CLPS	P11148	Colipase (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.79	SLN	F1P0F7	Sarcoplipin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.82	BLACAT1	A0ABV1AD43	Bladder cancer associated transcript 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.93	RPN2	P80897	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 2	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.06	C12orf75	A0A452J810	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.08		P83145	Ribonuclease UK114	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.13	CRABP2	P30370	Cellular retinoic acid-binding protein 2 (Fragment)	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.24	PYY	P29203	Peptide YY-like	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.26		A0ABV0ZFU7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.48		P80389	Antimicrobial peptide CHP1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.50		A0ABV0X2M7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.52		A0A3Q3AA92	Protein phosphatase 4 catalytic subunit	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.66		A0ABV0XFW2	Senescence-associated protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.88		A0ABV0XVD9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.91		A0ABV0XA39	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.99		A0ABV0X2W0	Cation-transporting P-type ATPase N-terminal domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01		A0ABV0XDA0	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02		A0ABV1A172	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.24	TMSB4B	A0A3Q2UAP0	Thymosin beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.31		A0ABV0Z9D2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.33		A0ABV0ZBY8	TCL1 upstream neural differentiation-associated RNA	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.37		A0ABV1ABK8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.43		A0ABV0YLA8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.51		A0ABV0YTH9	WAP domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.53	C7orf73	A0A1D5PNN4	Short transmembrane mitochondrial protein 1	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.57		A0ABV0XA75	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.64		A0ABV0X4T1	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.68	COX14	A0ABV1A933	COX14, cytochrome c oxidase assembly factor	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.70		A0ABV0XSF1	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.73		A0ABV0Y3E2	MICOS complex subunit MIC10	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.73		A0ABV0X8B6	KRAB domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.77		A0ABV0ZM25	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.81	PIGBOS1	A0ABV0ZSH1	PIGB opposite strand 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.81		A0ABV0YKP7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.94	PMAIP1	A0A1D5PAR2	Phorbol-12-myristate-13-acetate-induced protein 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.96		A0ABV0ZKX6	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.97	DNAJB8	A0ABV1A959	DnaJ heat shock protein family (Hsp40) member B8	4.0-6.0 kDa	Heat shock proteins	configured audit family match: Heat shock proteins
5.98		A0ABV0Z7P7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
6.03		P85000	Trypsin inhibitor CITI-1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.04		P0C915	Overexpressed in colon carcinoma 1 protein homolog	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.05		A0ABV1AFL1	THAP-type domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09	PLN	P26677	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09	PLN	R4GK80	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.11	TOMM5	A0ABV0X9V5	Translocase of outer mitochondrial membrane 5	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.12		A0ABV0XPE6	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.12		A0ABV0WZT7	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.18		A0ABV1A686	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.21	COPS9	A0A3Q2UDI0	COP9 signalosome complex subunit 9	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.22		A0ABV0XV54	HCV F-transactivated protein 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.24	C14orf2	A0A3Q2UCU2	6.8 kDa mitochondrial proteolipid	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.24	LOC107055098	A0A3Q2UFR1	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.24		A0ABV0X5W2	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.28	MT3	A4PBT0	Metallothionein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.29		A0ABV0X3L5	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.32	TOMM7	A0A1L1RYN1	Mitochondrial import receptor subunit TOM7 homolog	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.32	UQCR11	E1BXT9	Ubiquinol-cytochrome c reductase, complex III subunit XI	6.0-8.0 kDa		configured audit family match: Mitochondrial markers
6.37		A0ABV1A3S1	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.37		A0ABV0X5S1	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
6.38	MT-ATP8	P14093	ATP synthase F(0) complex subunit 8	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.38	RPL39	Q98TF5	Large ribosomal subunit protein eL39	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.41	SMIM3	A0A8V0Z3W9	Small integral membrane protein 3	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.41		A0A8V0ZP14	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.45		A0A8V0YUW4	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.46		P68497	Metallothionein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.48		A0A8V0Z473	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.48		A0A8V0Z9K5	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.49		P40667	SRY-related protein CH3 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.50	LOC107051594	A0A3Q2U1J3	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.52		P40666	SRY-related protein CH2 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.53		P40669	SRY-related protein CH7 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.53	ATP5E	A0A8V0YNP2	ATP5E synthase	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.55		P40670	SRY-related protein CH31 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.55		P40668	SRY-related protein CH4 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.56		A0A8V0X8T5	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.56		P40671	SRY-related protein CH32 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.57		P40665	SRY-related protein CH1 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.58	ENHO	A0A1D5NVP7	Energy homeostasis associated	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.58		A0A8V0X5X3	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.60	GAL14	Q0E4V3	Gallinacin-14	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.61		A0A8V1ADN4	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.61		A0A8V0XRF8	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.63		A0A8V0XED2	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.64		A0A8V0XC61	Carnitine O-palmitoyltransferase N-terminal domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.65		A0A8V0ZE77	Cortexin domain containing 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.69		A0A8V0XTC8	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.73		A0A8V1AHN9	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.74	SERF2	A0A3Q2ULA2	Small EDRK-rich factor 2	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.77	SMCO4	R4GG67	Single-pass membrane and coiled-coil domain-containing protein 4	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.77	PCP4	A0A8V0YKG6	Purkinje cell protein 4	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.88		A0A8V0Z175	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.89		A0A8V0ZT72	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.89	LOC107049692	A0A1D5P0A3	Annexin	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.89		A0A8V0XLS7	Rlla domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.90		A0A8V0YBT2	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.91		A0A8V0X2R4	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.92		A0A3Q2U1C9	Annexin	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.92		A0A8V0XZ16	Annexin	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.97		A0A3Q2U6A4	Annexin	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.97		A0A3Q3AXN8	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
7.01		A0A1D5P8E9	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38 δ western blot	40.0-42.0 kDa	ORI reported p38 δ bands were compiled from a source image representing β -actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38 δ -labeled bands were compiled from a β -actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_Glut4	Nrf2, NF-kB p65, PPAR α / δ , Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	chicken_UP000000539_9031.fasta
Marker file	chicken_marker_genes.tsv
Signaling-prefix file	chicken_signaling_prefixes.tsv
Family-pattern file	chicken_family_patterns.tsv
Species/report label	Gallus gallus